Designing a More Personalized Music Discovery Experience for College Students

A sponsored capstone project exploring how Spotify could make music discovery feel more playful, personal, and useful on web.

For my senior capstone, my team worked on a sponsored project focused on Spotify and college students' audio streaming habits.

The project began with a broad problem space: understanding how college students listen to music, discover new audio, and use Spotify across different contexts. Through exploratory research, personas, journey mapping, concept testing, prototyping, and usability testing, we narrowed the opportunity to music discovery.

Our final concept, **Discover**, was a web-based Spotify experience designed to help users explore music in a more personalized and playful way, then curate playlists that fit their current mood or listening needs.

Project Snapshot

Project / Client: Spotify sponsored capstone project
Timeline: September 2021 – June 2022
Role: Student UX Researcher
Team / Stakeholders: Student capstone team, Spotify research partners, faculty advisors
Problem Space: Understanding college students' audio streaming habits and identifying an opportunity for Spotify to increase visibility with its target market on web
Methods: Exploratory research, user interviews, contextual inquiry, personas, journey mapping, storyboarding, concept testing, A/B usability testing, prototyping
Tools: Figma, Dscout, Miro, Google Workspace
Outputs: Personas, journey maps, storyboards, concept test findings, A/B usability findings, high-fidelity web prototype, final poster, final video reel, and final presentation
Outcome / Impact: Designed "Discover," a personalized web-based music discovery concept; project received the Award of Excellence for Research

Focus Areas: Product discovery, music streaming, personalization, concept testing, A/B testing, web experience

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The Challenge

Spotify already has a strong mobile presence and a history of innovative music discovery features. Our challenge was to identify an opportunity for Spotify to better connect with college students, especially through the web experience.

Our initial problem space was:

> **Understand college students and their audio streaming habits and experiences to find an opportunity for Spotify to increase visibility to its target market on the web.**

As our research progressed, we narrowed the focus to music discovery.

Our refined design question became:

> **How might we create a music discovery process that is more personalized, playful, and builds on Spotify's history of innovative features?**

My Role

As a student UX researcher, I contributed to the full research and design process.

My work included exploratory research, user interviews, contextual inquiry, synthesis, persona development, journey mapping, storyboarding, concept testing, and A/B usability testing. I also helped translate research findings into design priorities and prototype iterations.

Because this was a sponsored capstone, our team also had to present our thinking clearly to external stakeholders and explain how research shaped the final design direction.

Research Approach

We began with exploratory research to understand how college students use audio streaming in everyday life.

We wanted to understand:

When and why students listen to music
How they discover new songs or artists
What role music plays in studying, socializing, commuting, relaxing, or getting ready
Where Spotify’s existing experience supported discovery well
Where discovery felt passive, repetitive, or difficult

Our exploratory research revealed three important themes:

Social listening — some users wanted more social connection through music, while others preferred a more private experience.
Discovery friction — users often found discovering new music difficult.
Need-based listening — users frequently listened to music for a specific purpose, mood, or activity.

Personas

Based on our research, we developed three personas to represent different listening behaviors:

The Curator

This user enjoys organizing, saving, and shaping their own music experience. They care about control and intentionality.

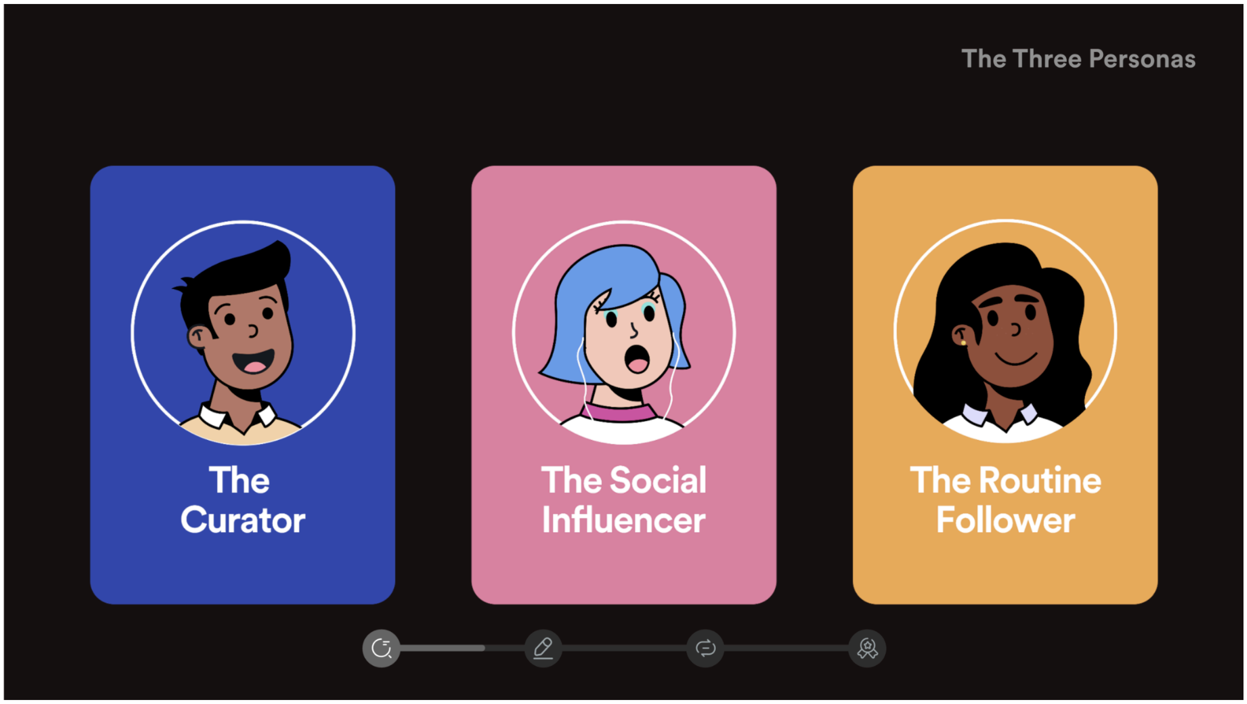
The Social Influencer

This user sees music as a form of social expression and connection. They may care about sharing, trends, and how music shows up in social settings.

The Routine Follower

This user relies on familiar listening patterns and may return to the same playlists, genres, or moods depending on context.

These personas helped us avoid designing for one generic “college student” and instead think about different motivations behind listening and discovery.

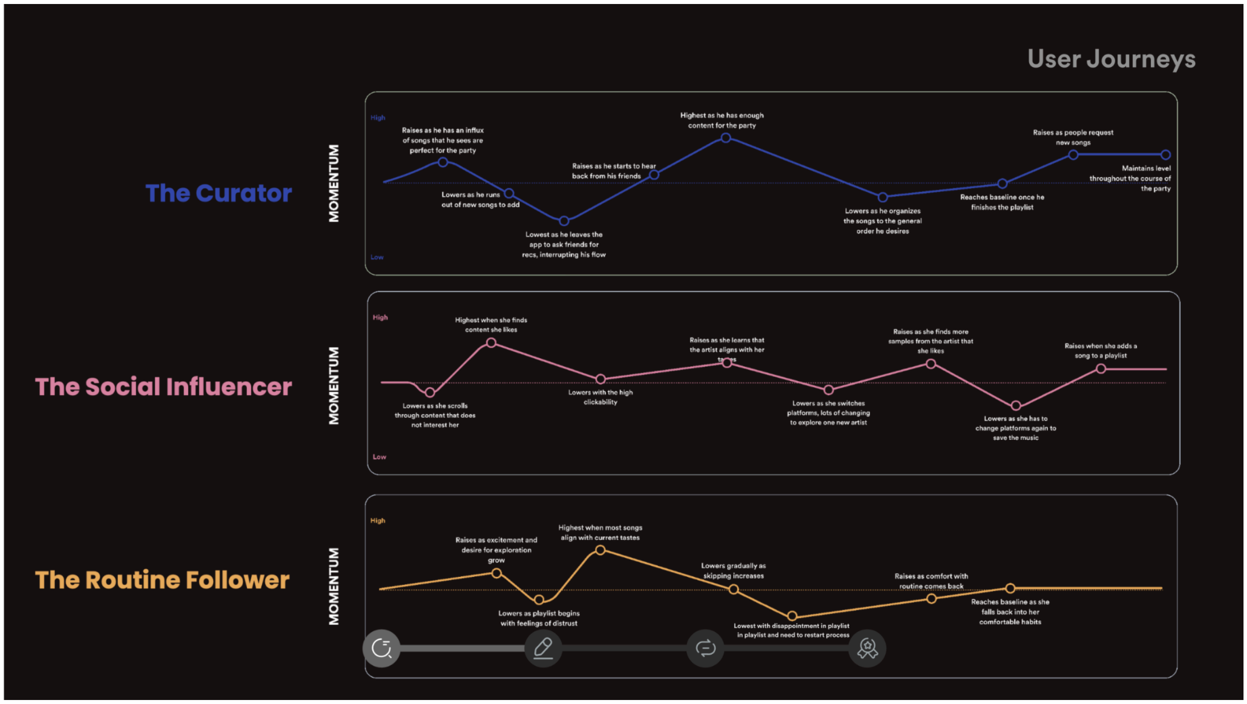


Journey Mapping

We created journey maps to understand how each persona moved through a music discovery experience.

The maps helped us identify moments of momentum, friction, hesitation, and satisfaction. They also showed that discovery was not just about finding a new song. It was about whether the experience matched the user’s mood, purpose, and level of effort in that moment.

This helped us shift from “show users more music” to “help users discover music in a way that feels relevant and controllable.”

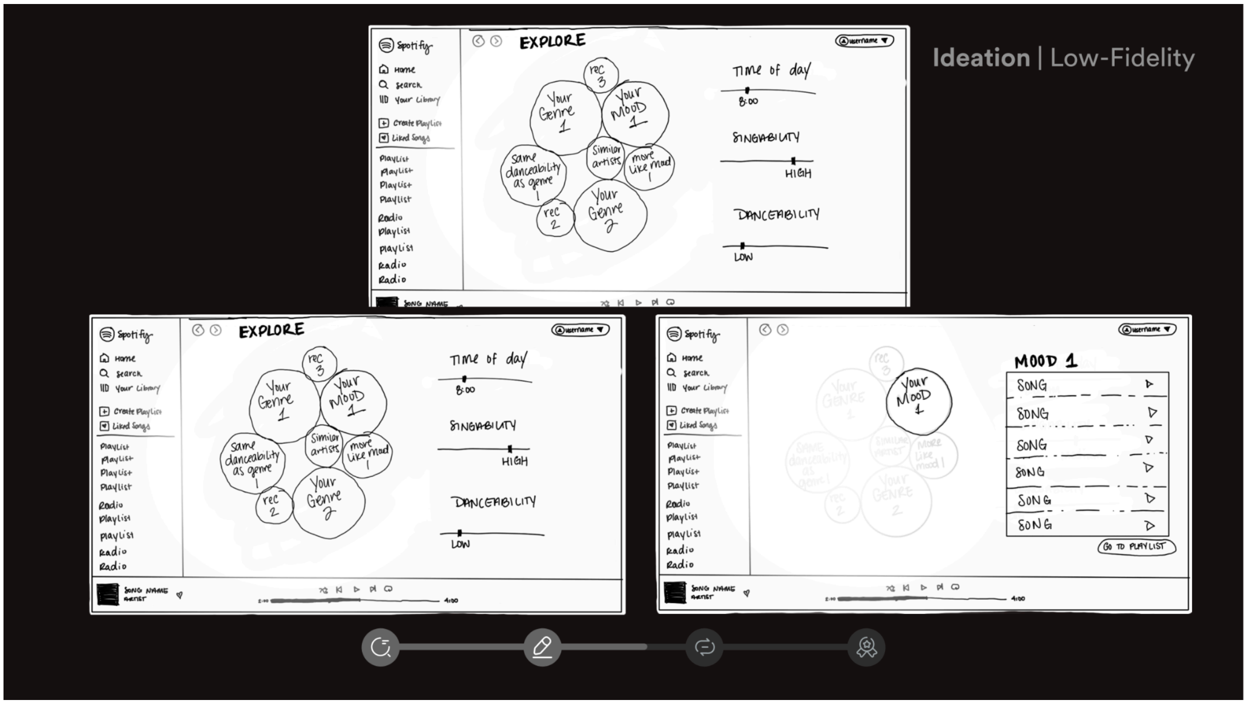


Ideation

During ideation, we explored multiple directions for improving Spotify's web discovery experience.

Some concepts were more globally oriented, helping users explore music trends across locations or cultures. Others focused more directly on mood, routine, genre, and playlist customization.

We were drawn to concepts that made discovery feel more active and playful while still helping users arrive at a useful output: a playlist they actually wanted to listen to.



Concept Testing

We used Dscout to test the concept with participants through 45-minute sessions.

From concept testing, we learned that participants:

Liked the concept and felt it achieved its objective
Wanted more opportunities for customization
Wanted a more visually engaging experience with more color and shape

These findings helped us define four design priorities:

Playfulness
Engagement
Interactivity
Personalization

The research pushed us away from a static recommendation experience and toward something more tactile, expressive, and user-directed.

Prototyping

We created a medium-fidelity Figma prototype to explore how users could interact with genre, mood, and recommendation controls.

The experience centered on visual blocks representing different genres, moods, or recommendation categories. Users could explore, adjust filters, and generate a playlist based on their preferences.

The goal was to make discovery feel less like scrolling through a list and more like shaping a playlist through interaction.

A/B Usability Testing

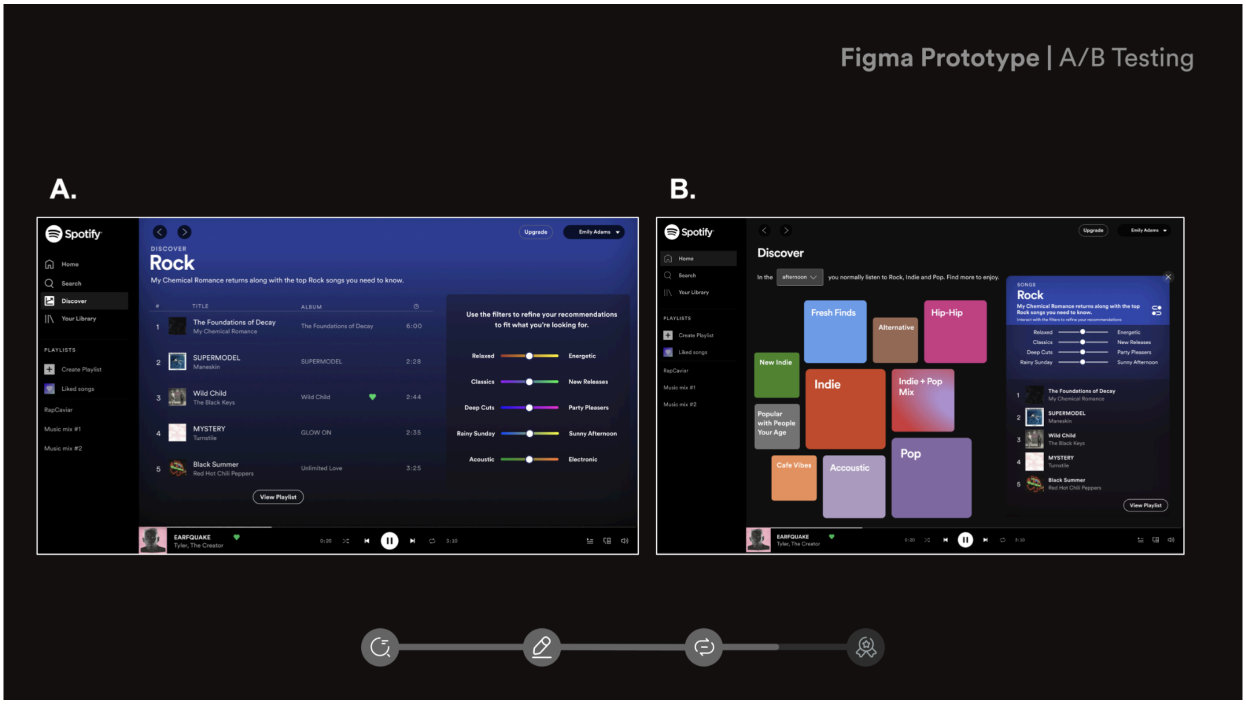
We tested two prototype directions to understand which interaction model better supported discovery and playlist creation.

The A/B test compared different ways of presenting recommendations, filters, and genre exploration.

Testing helped us refine the final experience by clarifying:

How users expected to interact with genre and mood controls
Which layout made discovery feel more intuitive

Where users wanted more customization
How the experience could better support playlist generation



Final Concept: Discover

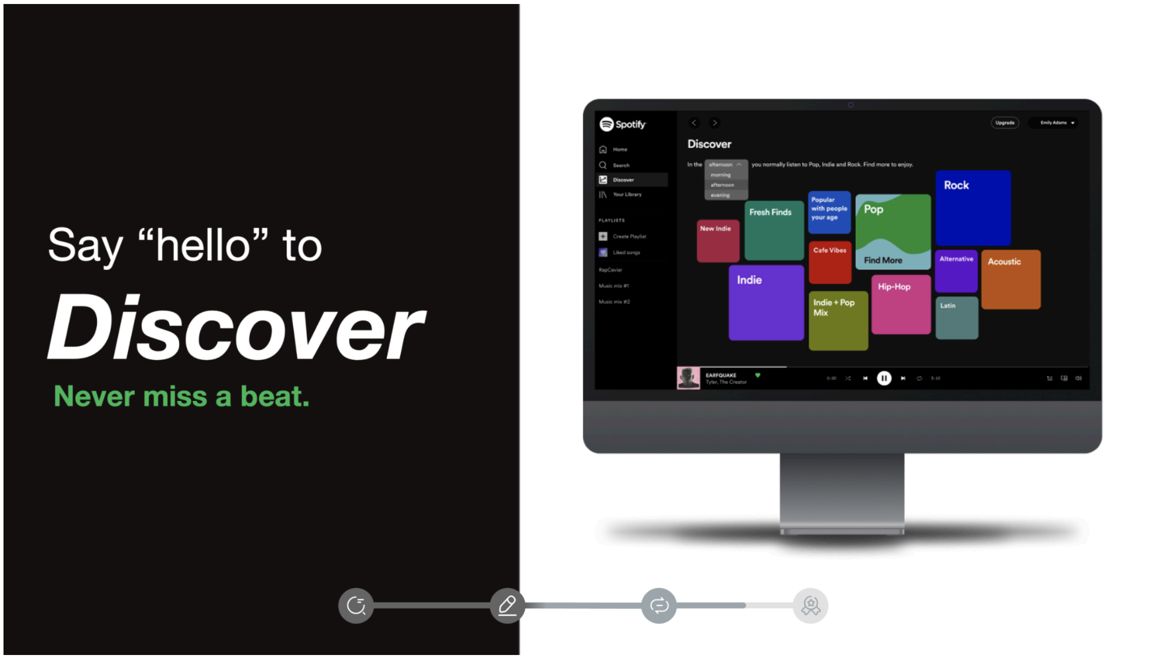
Our final concept was **Discover**, a web-based Spotify experience designed to help users discover and curate a playlist that aligned with their current mood.

The experience included:

A dedicated discovery page
Visual genre and mood blocks
Interactive filtering controls
Playlist generation
A playful visual system that supported exploration
A more personalized discovery flow

The core goal was:

> **Discover and curate a playlist that aligns well with the user's current mood.**



Key Feature: Discovery Page

The Discovery Page invited users to explore music through visual categories rather than a traditional list.

Users could interact with different genre and mood blocks, see related recommendations, and begin shaping a playlist from the discovery space.

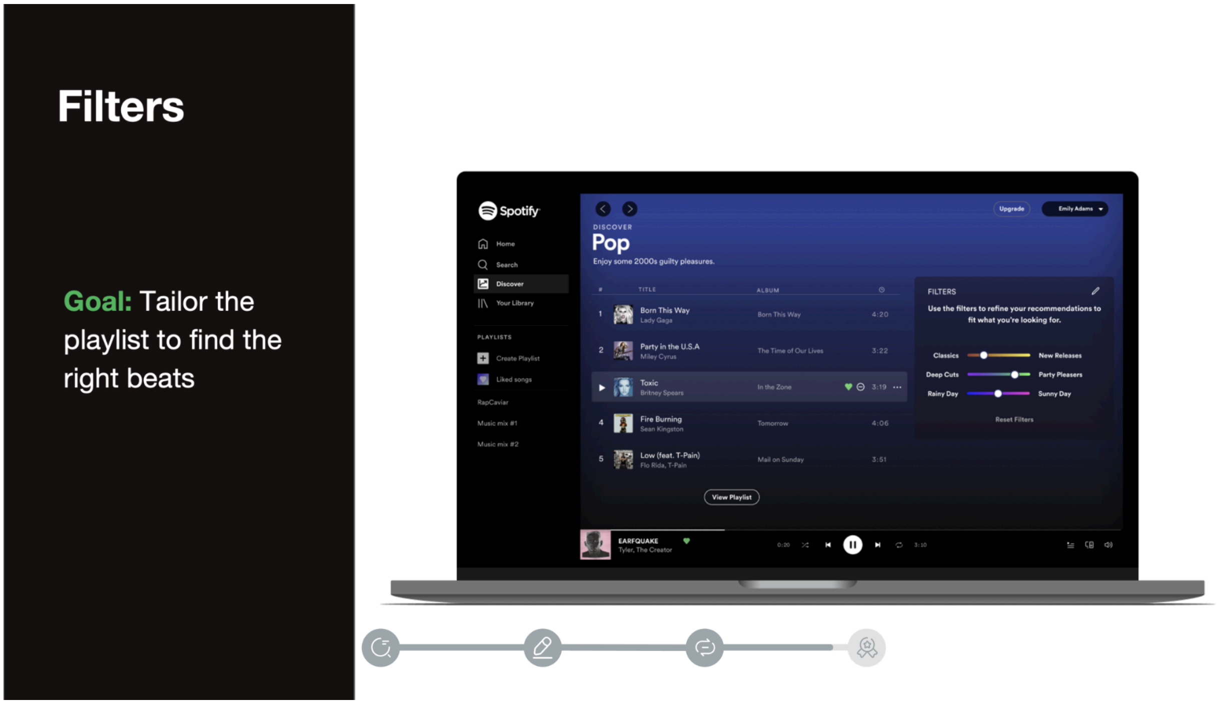
This helped make the process feel more active and exploratory.

Key Feature: Filters

The filter experience allowed users to tailor recommendations based on the kind of music they wanted.

Instead of assuming Spotify should know exactly what users wanted, the filters gave users more control over the recommendation process.

The goal was to help users “find the right beats” by adjusting the playlist toward their current mood or listening context.



Outcome

The project culminated in a final presentation, prototype, poster, and video reel presented at the Capstone Showcase.

The project received the **Award of Excellence for Research**, recognizing the strength of our research process and how it shaped the final product direction.

Reflection

This project taught me how valuable exploratory research can be when the problem space is broad.

We started with a general question about college students and audio streaming habits, but research helped us narrow the scope to a clearer product opportunity: making discovery feel more personalized, playful, and interactive.

It also taught me that personalization is not just about algorithms. Users still want control, expression, and the ability to shape the experience themselves.

Looking back, this project was an important step in developing my ability to move from discovery research to concept testing, prototype iteration, and final product storytelling.

What I'd Do Differently Today

If I were approaching this project now, I would spend more time defining success metrics for the prototype.

For example, I would want to measure:

Whether users discovered music they were actually interested in
Whether users felt more in control of the recommendation process
Whether the interaction increased playlist saves or listening intent
How the experience compared to Spotify's existing discovery features

I would also test the experience with a larger and more diverse group of students to better understand how listening behaviors vary by context, identity, routine, and relationship to music.

Project Card

Designing a More Personalized Music Discovery Experience for College Students
Conducted exploratory research, personas, journey mapping, concept testing, and A/B usability testing for a sponsored Spotify capstone. Designed Discover, a web-based music discovery concept that helped users create playlists based on mood, genre, and listening context. Awarded Excellence for Research.

Tags

UX Research · Product Discovery · Concept Testing · A/B Testing · Music Streaming · Personas · Journey Mapping · Storyboarding · Figma · Dscout · Prototyping · Web Experience · Personalization